

Executive Summary: Customer Churn Analysis

(Telecom Industry | 7,043 Records | Python Analysis)

1. Overview

This analysis investigates customer churn using a dataset of **7,043 customer records** from a telecommunications company. With **21 variables** spanning demographics, service subscriptions, billing details, and churn status, the study uses Python tools (*Pandas, NumPy, Matplotlib, Seaborn*) to uncover patterns and visualize insights—highlighting a key factor: **payment method significantly influences churn rates**.

2. Data Preprocessing

a. Data Loading & Structure

- **Source:** Customer churn.csv
- **Features:**
 - 18 object columns (e.g., gender, Contract, Churn)
 - 2 integer columns (SeniorCitizen, tenure)
 - 1 float column (MonthlyCharges)
- **Issue:** TotalCharges incorrectly stored as object due to blanks.

b. Cleaning & Transformation

- Blank TotalCharges entries filled with 0 → converted to float.
- No missing values (`df.isnull().sum() = 0`)
- SeniorCitizen: Transformed from 0/1 to "No"/"Yes" → 16.21% are seniors.
- No duplicate customerID entries found.

c. Descriptive Statistics

- **Tenure:** Avg. 32.37 months (0–72 range)
- **Monthly Charges:** Avg. \$64.76
- **Total Charges:** Avg. \$2,279.73

- **Senior Citizens:** 16.21% of total customers
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3. Key Findings [🔍]

a. Overall Churn Rate [📊]

- Total Churned: **1,869 (26.54%)**
- Active Customers: **5,174 (73.46%)**

b. Churn by Payment Method [📊]

- Four payment methods were analyzed:
 - **Electronic Check:**
 - **Highest churn rate: 45.41%**
 - 57.46% of all churned users
 - **Mailed Check:**
 - 19.11% churn rate
 - **Bank Transfer (Auto):**
 - 16.73% churn rate
 - **Credit Card (Auto):**
 - **Lowest churn rate: 15.03%**

Insight: Manual payment users (especially Electronic Check) show **significantly higher churn**. Automatic methods correlate with **greater customer retention**.

4. Visualizations [📊]

Bar Chart: Churn by Payment Method

- Design: Large size, clear fonts, color-coded by churn status
 - X-axis: Payment Method (rotated labels)
 - Y-axis: Customer count
 - **Highlight:**
 - **Electronic Check users** make up **57.46%** of all churns
 - Visual cue for targeting **retention efforts**
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5. Conclusions & Recommendations [✓]

a. Key Takeaways

- **Electronic Check** users are at high churn risk
- **Automated payment** users show **stability and satisfaction**
- Contract type and tenure may further explain churn behavior

b. Strategic Recommendations [🚀]

1. **Improve Electronic Check UX**
 - Simplify process, reduce errors, build trust.
 2. **Incentivize Auto-Payment Adoption**
 - Offer discounts or benefits to switchers.
 3. **Proactive Customer Support**
 - Target high-risk groups with helpdesk & education.
 4. **Deeper Feature Interaction Analysis**
 - Explore churn by tenure, contract type, and service bundles.
 5. **KPI Monitoring**
 - E.g., Reduce churn from Electronic Checks by **10% in 6 months**
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6. Next Steps [🧭]

- **Expand EDA:** Include contract type, tenure, service use
 - **Machine Learning:** Build churn prediction models
 - **Cross-Team Collaboration:** Present insights to billing & marketing
 - **Customer Feedback Loop:** Run surveys with Electronic Check users
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Artifact: Summary Document [📄]

This executive summary and accompanying visualizations are designed for **stakeholder presentations**, highlighting **actionable insights and measurable goals** for improving retention and operational efficiency.